

0.0.1 Learning Bound for Finite Hypothesis Class — Consistent Case

Let $\mathcal{H}$ be a finite hypothesis class mapping $\mathcal{X} \rightarrow \mathcal{Y}$, and let $\mathcal{A}$ be an algorithm that, for any target concept $c \in \mathcal{H}$ and i.i.d. sample S , returns a hypothesis h_S that is consistent with S ; that is, $L_{S,c}(h_S) = 0$. Then, for any $\epsilon, \delta > 0$, the following holds:

$$\mathbb{P}_{S \sim \mathcal{D}^m} [L_{\mathcal{D},c}(h_S) \leq \epsilon] \geq 1 - \delta$$

provided that

$$m \geq \frac{1}{\epsilon} \left(\log \left(\frac{1}{\delta} \right) + \log |\mathcal{H}| \right).$$

This yields an equivalent generalization bound: for any $\epsilon, \delta > 0$, with probability at least $1 - \delta$,

$$L_{\mathcal{D},c}(h_S) \leq \frac{1}{m} \left(\log |\mathcal{H}| + \log \left(\frac{1}{\delta} \right) \right).$$

0.0.2 Learning Bound for Finite Hypothesis Class — Non-Consistent Case

Let $\mathcal{H}$ be a finite hypothesis class. Then, for any $\delta > 0$, with probability at least $1 - \delta$, the following inequality holds uniformly for all $h \in \mathcal{H}$:

$$L_{\mathcal{D},c}(h) \leq L_{S,c}(h) + \sqrt{\frac{1}{2m} \left(\log \left(\frac{2}{\delta} \right) + \log |\mathcal{H}| \right)}.$$

0.0.3 See also

0.0.4 References